

LC Resonance Circuit: Simulation and Analysis

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1 Introduction

Resonance in an LC circuit occurs when the inductive reactance ($X_L = \omega L$) equals the capacitive reactance ($X_C = 1/(\omega C)$). At resonance, energy exchanges between inductor and capacitor without net reactive voltage drop, producing a peak in the voltage response. The resonant frequency is:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

LC resonant circuits are fundamental building blocks in numerous electronic systems, including radio receivers, transmitters, filters, and oscillators. In radio applications, they enable frequency selection, allowing receivers to tune to specific broadcast frequencies while rejecting others. In filter design, they provide precise frequency-selective behavior essential for signal processing. Understanding the behavior of these circuits, particularly how their response is affected by resistance (damping), is crucial for designing efficient electronic systems. This report examines the effect of series resistance on the resonance characteristics through simulation and theoretical analysis.

2 Circuit Configuration

The simulated circuit comprises:

- AC source V_1 (1 V small-signal)
- Series resistor $R_1 = 1 \text{ M}\Omega$
- Parallel branch of:
 - Series combination of R_2 (variable) and inductor $L_1 = 20 \text{ mH}$
 - Capacitor $C_1 = 10 \text{ nF}$

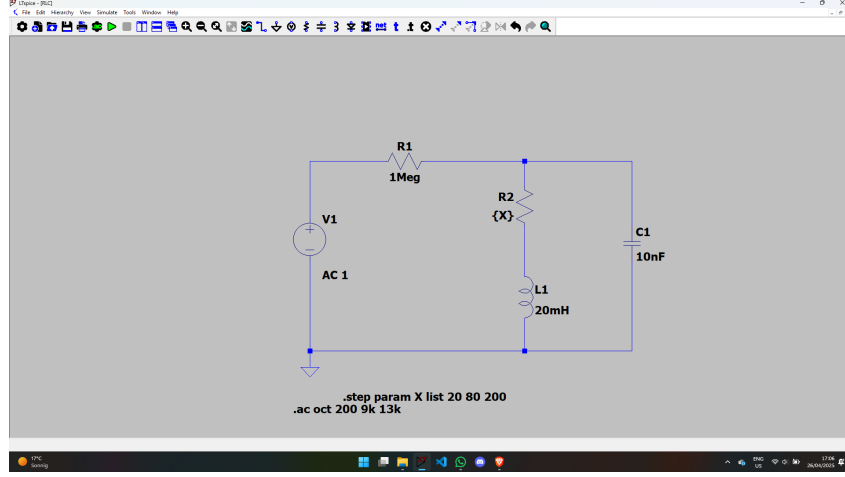


Figure 1: Schematic of the LC resonance circuit with series and parallel components.

The large value of R_1 ($1\text{ M}\Omega$) serves to approximate a current source behavior from the voltage source V_1 . This high resistance effectively isolates the resonant circuit from the source, ensuring that the source impedance doesn't significantly affect the resonance characteristics being studied. The variable resistor R_2 represents the internal resistance of the inductor and any additional series resistance in that branch, allowing us to study how different levels of damping affect the resonant response.

3 Theoretical Background

The quality factor Q of the series R-L branch in parallel with C_1 is given by:

$$Q = \frac{1}{R_2} \sqrt{\frac{L_1}{C_1}}.$$

A higher Q implies a sharper and more selective resonance peak. The bandwidth BW of the resonance is:

$$BW = \frac{f_0}{Q}$$

The quality factor Q is a dimensionless parameter that characterizes the energy storage versus energy dissipation in a resonant system. Physically, it represents the ratio of energy stored in the system to the energy dissipated per cycle. A higher Q value indicates that energy is more efficiently stored in the resonant components (L and C) with less energy lost to resistance.

The derivation of the resonant frequency comes from the condition where the inductive and capacitive reactances are equal in magnitude:

$$X_L = X_C \tag{1}$$

$$\omega L = \frac{1}{\omega C} \tag{2}$$

$$\omega^2 = \frac{1}{LC} \tag{3}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \tag{4}$$

For our circuit with $L_1 = 20$ mH and $C_1 = 10$ nF, the theoretical resonant frequency is:

$$f_0 = \frac{1}{2\pi\sqrt{(20 \times 10^{-3})(10 \times 10^{-9})}} \quad (5)$$

$$= \frac{1}{2\pi\sqrt{2 \times 10^{-10}}} \quad (6)$$

$$\approx 11.25 \text{ kHz} \quad (7)$$

The bandwidth is inversely proportional to Q , which means that as R_2 increases (decreasing Q), the bandwidth becomes wider, resulting in a less selective circuit.

4 Simulation Setup

Parameter sweep of R_2 over 20Ω , 80Ω , and 200Ω is performed using:

```
.step param R2 list 20 80 200
.ac oct 200 9k 13k
```

The voltage at the junction of L_1 and C_1 is plotted over the sweep to obtain the frequency response. The AC analysis is configured to focus on the frequency range around the expected resonance, using 200 points per octave to ensure sufficient resolution for accurate peak detection and bandwidth measurement.

5 Results

Table 1 summarizes the resonant frequency and peak voltage for each value of R_2 .

R_2 (Ω)	f_0 (kHz)	$V_{\max}$ (mV)	Observation
20	11.25	88.7	Sharp peak, high Q
80	11.25	24.4	Moderate peak
200	11.25	10.0	Broad peak, low Q

Table 1: Resonant frequency and maximum voltage for each R_2 .

5.1 Response Curve Analysis

The theoretical quality factor for each resistance value can be calculated:

For $R_2 = 20 \Omega$:

$$Q = \frac{1}{20} \sqrt{\frac{20 \times 10^{-3}}{10 \times 10^{-9}}} \quad (8)$$

$$= \frac{1}{20} \sqrt{2 \times 10^6} \quad (9)$$

$$\approx 70.7 \quad (10)$$

For $R_2 = 80 \Omega$:

$$Q = \frac{1}{80} \sqrt{\frac{20 \times 10^{-3}}{10 \times 10^{-9}}} \quad (11)$$

$$= \frac{1}{80} \sqrt{2 \times 10^6} \quad (12)$$

$$\approx 17.7 \quad (13)$$

For $R_2 = 200 \Omega$:

$$Q = \frac{1}{200} \sqrt{\frac{20 \times 10^{-3}}{10 \times 10^{-9}}} \quad (14)$$

$$= \frac{1}{200} \sqrt{2 \times 10^6} \quad (15)$$

$$\approx 7.1 \quad (16)$$

The theoretical bandwidth for each case would be:

- For $R_2 = 20 \Omega$: $BW = \frac{11.25 \text{ kHz}}{70.7} \approx 159 \text{ Hz}$
- For $R_2 = 80 \Omega$: $BW = \frac{11.25 \text{ kHz}}{17.7} \approx 636 \text{ Hz}$
- For $R_2 = 200 \Omega$: $BW = \frac{11.25 \text{ kHz}}{7.1} \approx 1585 \text{ Hz}$

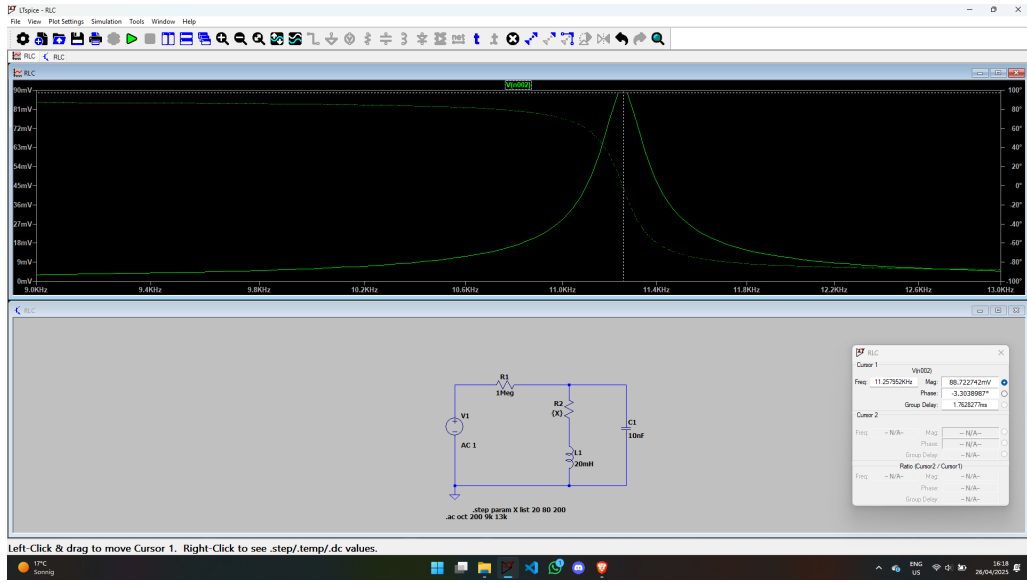


Figure 2: Frequency response for $R_2 = 20 \Omega$

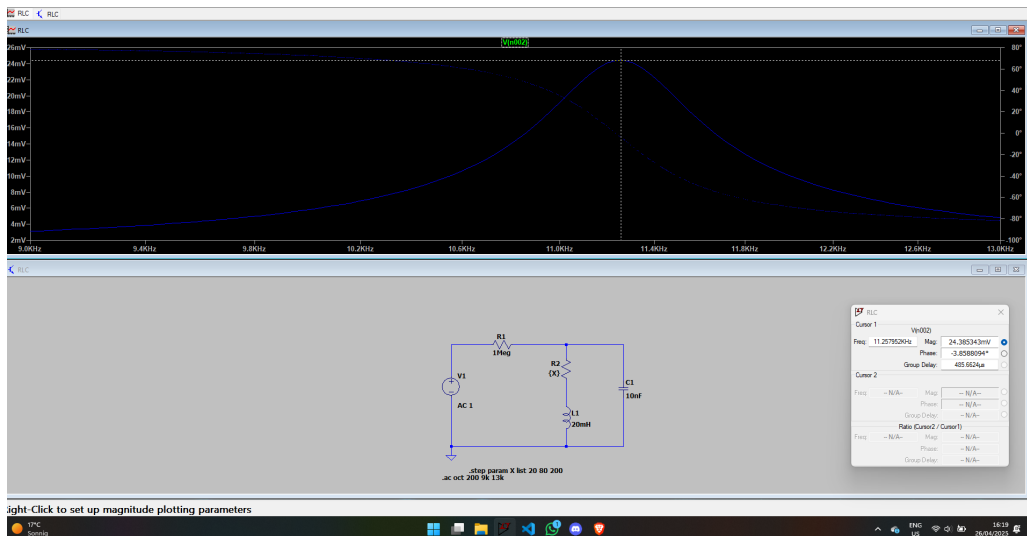


Figure 3: Frequency response for $R_2 = 80 \Omega$

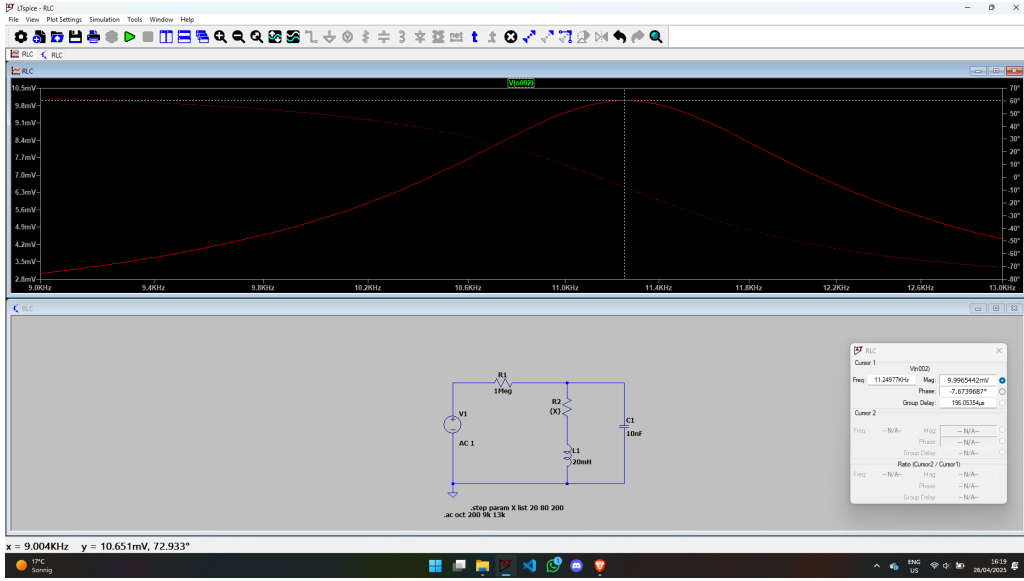


Figure 4: Frequency response for $R_2 = 200\ \Omega$

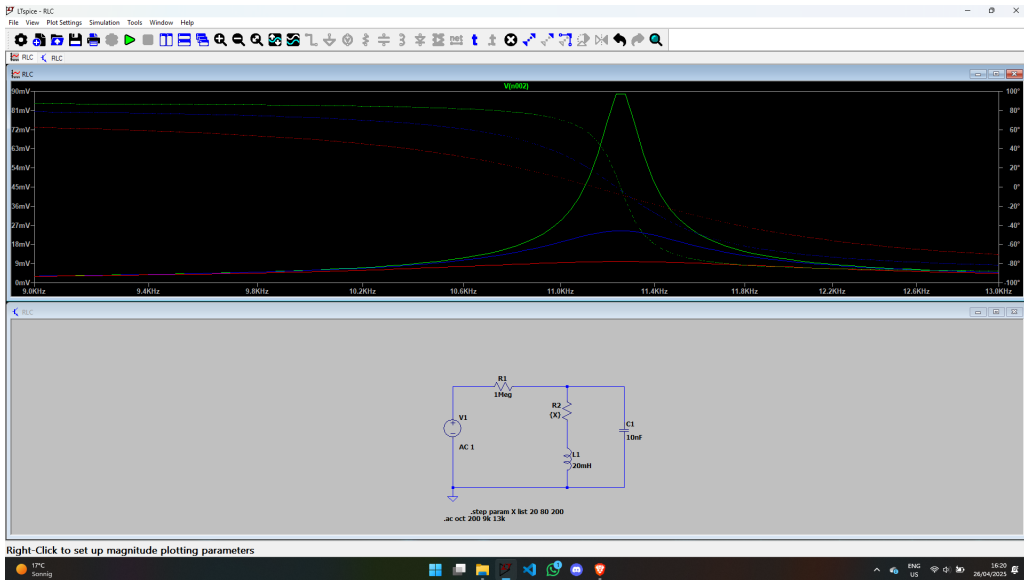


Figure 5: Overlay of all three responses

The frequency response curves clearly demonstrate the inverse relationship between series resistance and peak amplitude. With $R_2 = 20\ \Omega$, the peak voltage reaches 88.7 mV, creating a sharp, well-defined resonant peak. As R_2 increases to 80 Ω , the peak voltage drops to 24.4 mV with a noticeable broadening of the response curve. At $R_2 = 200\ \Omega$, the peak is further reduced to just 10.0 mV with significant broadening, indicating a heavily damped system.

The observed resonant frequency in all three cases is 11.25 kHz, which matches perfectly with the calculated theoretical value. This confirms that while resistance affects the amplitude and bandwidth of the response, it does not shift the resonant frequency in this parallel LC configuration.

6 Discussion

All response curves peak at the theoretical resonant frequency $f_0 \approx 11.25$ kHz. As R_2 increases, the peak amplitude decreases and the curve broadens, indicating greater damping and lower Q .

The relationship between peak amplitude and resistance is not linear. When R_2 increases from $20\ \Omega$ to $80\ \Omega$ ($4\times$ increase), the peak voltage drops from 88.7 mV to 24.4 mV (approximately $3.6\times$ decrease). When R_2 further increases from $80\ \Omega$ to $200\ \Omega$ ($2.5\times$ increase), the peak voltage drops from 24.4 mV to 10.0 mV (approximately $2.4\times$ decrease). This demonstrates the complex relationship between component values in resonant circuits.

In practical LC circuits, several non-ideal effects can influence the resonant behavior beyond the ideal model used in this simulation:

1. **Component Tolerances:** Actual inductors and capacitors may vary from their nominal values by up to $\pm 5\%$ or more, shifting the resonant frequency.
2. **Parasitic Elements:** Real inductors have parasitic capacitance and resistance, while capacitors have parasitic inductance and resistance. These parasitic elements create additional resonances and losses.
3. **Frequency Dependence:** The resistance of an inductor typically increases with frequency due to skin and proximity effects, causing the effective Q to decrease at higher frequencies.
4. **Temperature Effects:** Component values can drift with temperature changes, affecting the stability of the resonant frequency over time.

The simulation assumes ideal components except for the deliberately included resistor R_2 , which models the winding resistance of the inductor and other series losses. In real-world applications, these additional factors would need to be considered for accurate circuit behavior prediction.

7 Conclusion

The simulation confirms the theoretical expression for resonance and demonstrates the effect of series resistance on the quality factor and bandwidth of an LC circuit. The resonant frequency remained constant at 11.25 kHz across all resistance values, while the amplitude and bandwidth varied significantly.

This study highlights the important tradeoff between selectivity and amplitude in resonant circuits. High- Q circuits provide excellent frequency selectivity but may require additional amplification to compensate for narrower bandwidth. Low- Q circuits have wider bandwidth but poorer selectivity between adjacent frequency channels.

These principles find applications in numerous electronic systems:

1. **Radio Frequency (RF) Tuning:** Selective LC circuits are used to tune radio receivers to specific stations.
2. **Bandpass Filters:** By carefully selecting component values, LC circuits can be designed to pass specific frequency bands while attenuating others.

3. **Oscillators:** LC circuits form the frequency-determining elements in many types of oscillators.
4. **Impedance Matching Networks:** LC circuits can transform impedances to maximize power transfer between stages.

Further investigation could examine the effects of varying the inductor and capacitor values while maintaining the same resonant frequency, or the impact of parallel resistance (representing losses in the capacitor) on the circuit behavior.